# Artificial Intelligence Assignment # 2

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**Objective:** Design and implement a Strategy Engine using a Multi-Objective Genetic Algorithm (GA) to determine optimal deployment positions, sensor activation schedules, and waypoint trajectories for a fleet of 3 UAVs surveying a volatile 50x50 hazard grid while balancing coverage, energy, makespan, and safety constraints.

## 1. Design

The system operates across a 50x50 discretized grid containing high-priority disaster targets, environmental hazard pockets (radiation/wind), and restricted No-Fly Zones (NFZ). Unlike traditional rigid algorithms, our Strategy Engine employs a dynamic multi-segment chromosome to co-evolve flight routes, velocity, and sensor profiles simultaneously.

**• Chromosome Representation & Sub-Gene Encoding:**

Each candidate solution represents the entire fleet of 3 drones. Each drone has a dynamic-length sequence of waypoints $L \in [4, 12]$ to allow adaptive obstacle avoidance without redundant points.

| Gene Segment | Encoding Type | Values / State Space | Operational Trade-Off |
|---|---|---|---|
| Waypoint Trajectory | Continuous Coordinates (x,y) | x, y ∈ [0, 50], L ∈ [4, 12] | Path length vs disaster target coverage |
| Speed Profile | Discrete Real Vector | v ∈ {1.0, 2.0, 3.0} m/s | Flight time vs aerodynamic drag ($P \propto v^2$) |
| Sensor Power | Discrete Integer Vector | s ∈ {0 (0W), 1 (25W), 2 (60W)} | Detection radius (0, 3.5, 6.0) vs battery draw |

**• Multi-Objective Fitness & Physics Model:**

- **Propulsion Power:** $P_{\text{flight}}(v) = 20.0 + 7.5 \cdot v^2 \quad (\text{Watts})$
- **Sensor Power Stages:** $P_{\text{sensor}}(s) \in \{0\text{W } (s=0),\ 25\text{W } (s=1),\ 60\text{W } (s=2)\}$
- **Leg Energy & Makespan:** $E_{\text{leg}} = (P_{\text{flight}} + P_{\text{sensor}}) \cdot \frac{d_{\text{leg}}}{v_{\text{leg}}}, \quad T_{\text{mission}} = \max_d \left(\sum \frac{d_{\text{leg}}}{v_{\text{leg}}}\right)$
- **Dynamic Hard Penalties:** $\text{Penalty}_{\text{NFZ}} = 90.0 \cdot N_{\text{blocked}}, \quad \text{Penalty}_{\text{Battery}} = 0.03 \cdot \max(0, E_d - 18{,}000)$
- **Multi-Objective Fitness:** $\textbf{Fitness} = 3.5 \cdot \text{Coverage} - 0.004 \cdot E_{\text{total}} - 0.4 \cdot T - 2.2 \cdot \text{Hazard} - \textbf{Penalties}$

## 2. Genetic Operators & Constraint Handling

To explore the spatial solution space effectively without destroying valid route structures, custom genetic operators were developed:

- **Selection:** Tournament selection with size $k = 4$ balances selection pressure and genetic diversity.
  - **Dynamic Cut-and-Splice Crossover (p = 0.85):** Trajectories of unequal lengths are cut at random indices and spliced together, clamping child lengths between 4 and 12 waypoints. Fleet route-swapping occurs at $p = 0.30$.
  - **Dynamic Insertion Mutation:** Adds an intermediate waypoint with Gaussian perturbation to synthesize obstacle detours.
  - **Dynamic Deletion Mutation:** Removes redundant waypoints to straighten flight lines and conserve energy.
  - **Spatial Coordinate Jitter (p = 0.08):** Shifts waypoint positions by $\Delta x, \Delta y \sim N(0, 2.5^2)$ to fine-tune coverage.
  - **Constraint Handling:** Dynamic penalty functions penalize invalid individuals rather than applying destructive repair operators, allowing the GA to learn obstacle avoidance naturally through survival of the fittest.

## 3. Execution & Evolved Flight Paths

The simulation executes over 70 generations with a population of 60 individuals. The 2D operational map below shows the final evolved trajectories for the 3 UAVs from base depot (2, 2) alongside active sensor scanning footprints.

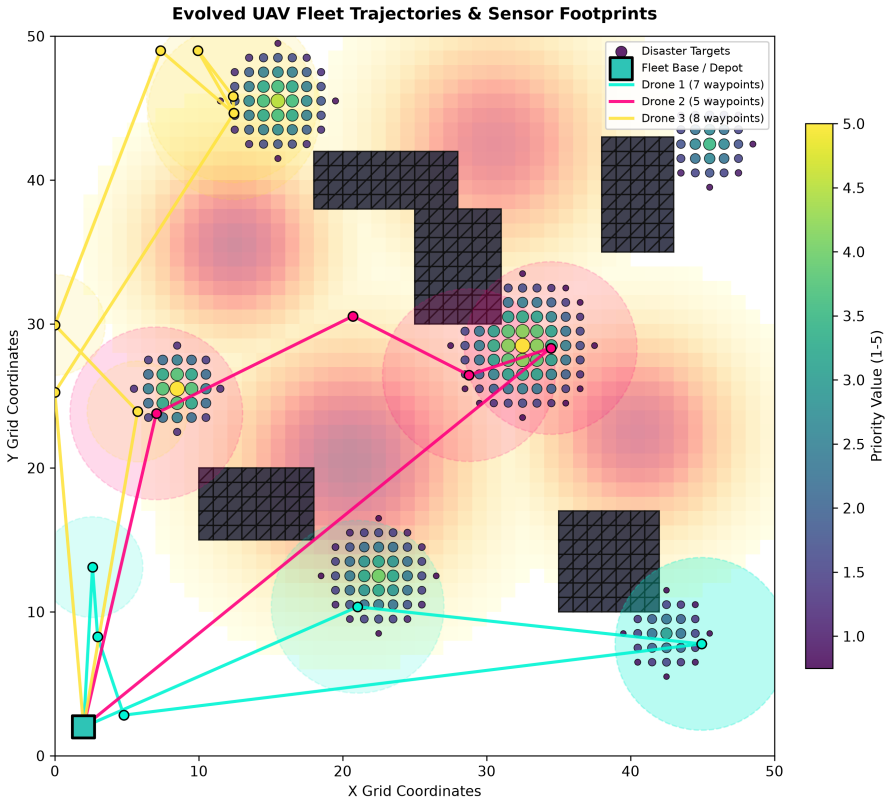



**Figure 1:** Evolved UAV flight trajectories on the 50x50 hazard grid. Hatched dark boxes represent restricted No-Fly Zones. Translucent circular footprints show active sensor scans covering the priority disaster clusters.

## 4. Output Visualizations & Convergence

The GA successfully converged toward an optimal mission profile. Below are the fitness convergence curves and generational evolution snapshots.

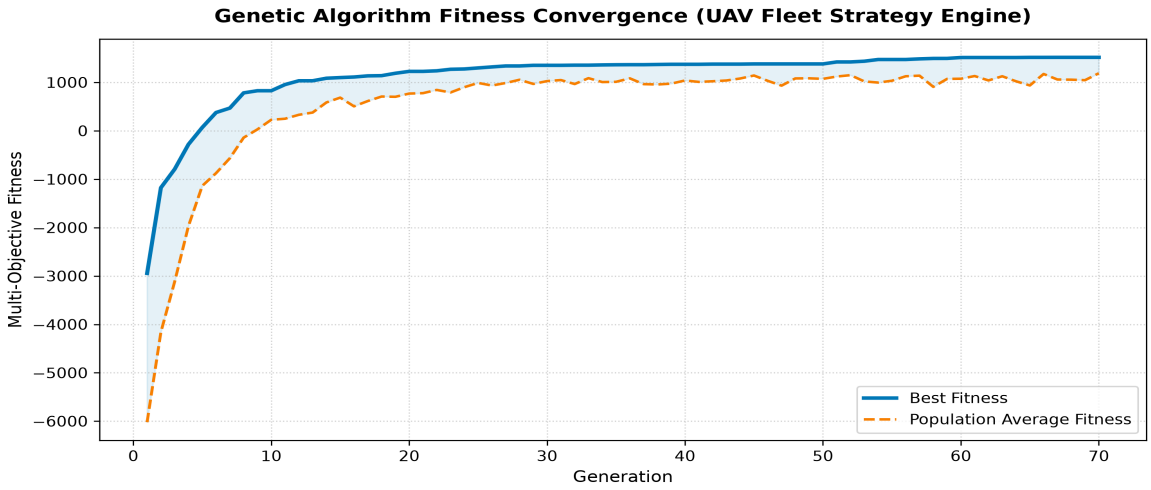



**Figure 2:** Fitness convergence across 70 generations. Best fitness climbs steadily from -2,945 (penalty zone) to +1,517 with zero constraint violations. Average fitness closely tracks best fitness, indicating maintained diversity.

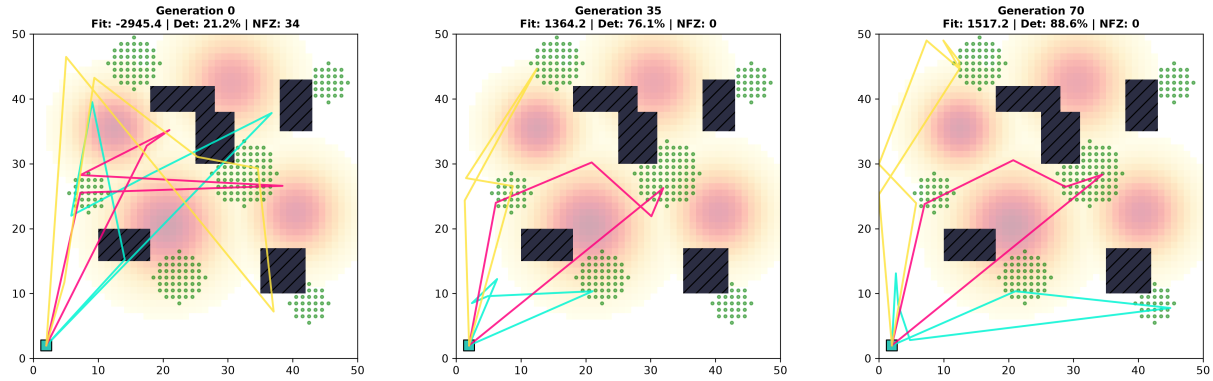
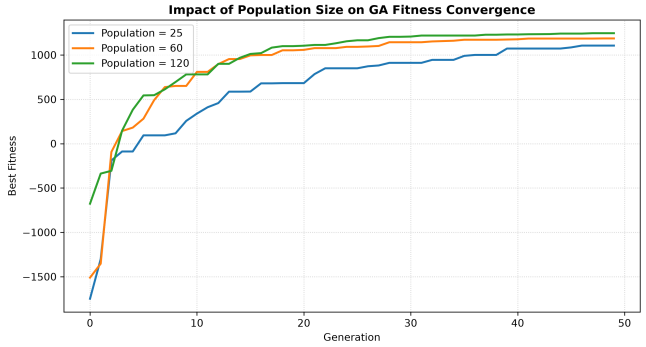



**Figure 3:** Visual evolution across Generation 0 (random routes, 34 NFZ violations, 21.2% detection), Generation 35 (intermediate, 0 violations, 76.1% detection), and Generation 70 (optimal mission plan, **88.6% detection rate**, 0 violations, 11,665 J total energy).

## 5. Hyperparameter Sensitivity Analysis

To evaluate the algorithmic behavior, we tested the impact of varying Population Size, Mutation Rate, and Selection Pressure.



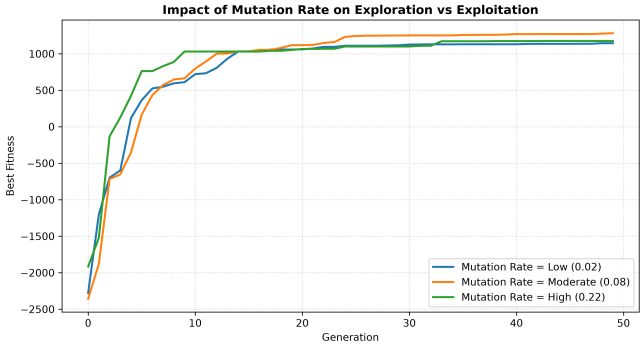




**Figure 4 (Left): Population Size (25, 60, 120):** Population 25 suffers from premature convergence (plateau at 1,105) due to limited diversity. Population 60 provides steady evolutionary progress in 5.6s. Population 120 achieves a marginal improvement (1,248) at the expense of double the computational time.
**Figure 5 (Right): Mutation Rate (0.02, 0.08, 0.22):** A low rate (0.02) gets trapped in local optima (1,148). A high rate (0.22) disrupts good solutions through genetic noise (1,173). Moderate mutation (0.08) achieves optimal performance (1,282).

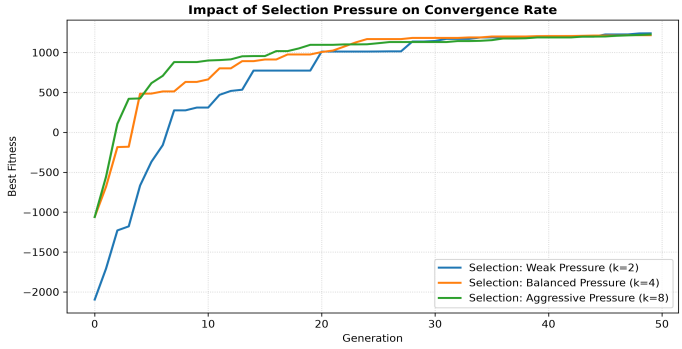



**Figure 6: Tournament Size (k = 2, 4, 8):**
• **k = 2 (Weak):** Converges slowly, taking >25 generations to clear initial collisions.
• **k = 4 (Balanced):** Smooth convergence with sustained population diversity.
• **k = 8 (Aggressive):** Rapidly clears violations by Gen 4, but loses diversity in late generations.
**Optimal Choice:** Tournament size k = 4 offers the best balance.

## 6. Comparative Study: GA vs. Baseline Random Search

We compared the evolved GA solution against a uniform Random Search baseline under an identical computational budget of 2,500 evaluations.

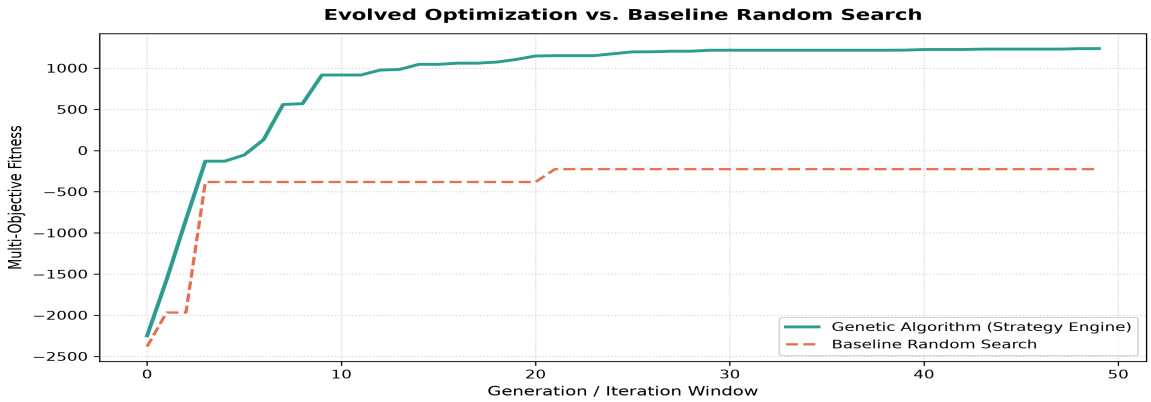



**Figure 7:** Fitness progression of the GA Strategy Engine vs Random Search over 2,500 candidate evaluations.

| Evaluation Metric | Evolved GA Solution | Baseline Random Search | Performance Delta |
|---|---|---|---|
| Multi-Objective Fitness | **+1,238.29** | -225.48 | **+1,463.77 pts (GA Superior)** |
| Disaster Target Coverage | **473.0 pts** | 221.8 pts | +113.3% higher coverage |
| Target Detection Rate | **89.0%** | 40.8% | +48.2% higher detection |
| No-Fly Zone (NFZ) Violations | **0 cells (100% Safe)** | 6 cells (Collisions) | Zero collision safety |
| Total Fleet Energy | **11,548.8 J** | 17,337.1 J | 33.4% energy conserved |
| Hazard Exposure | **49.88** | 132.52 | 62.4% risk reduction |

## 7. Discussion on Uniqueness of Evolved Strategy

The evolved GA strategy displays three key emergent behaviors that Random Search cannot replicate:
1. **Spatial Sector Partitioning:** The 3 UAVs divided the 50x50 space into distinct non-overlapping sectors (North, Central, and South).
2. **Dynamic Obstacle Circumnavigation:** The GA utilized dynamic waypoint insertion to route tightly around No-Fly Zones without crossing radiation pockets.
3. **Adaptive Sensor Throttling:** Sensor power was boosted to High-Power mode (60W) only when crossing dense survivor clusters and shut Off (0W) during the return leg to depot (2, 2).

**GitHub Repository:** Modular Python implementation with source code, visualizer, and execution instructions.